

BEFORE

From the advances of the 17th century, scientific progress gathered momentum, leading to many 19th-century innovations.

SCIENTIFIC REVOLUTION

During the 17th century, drastic changes took place as European science was revolutionized by the ideas of people like Galileo Galilei and Isaac Newton << 268–69. By the late 17th century, **dramatic advances** had been made in mathematics, astronomy, and biology.

INDUSTRIAL REVOLUTION

In the late 18th century, Western Europe began to industrialize rapidly << 292–95. Coal was used to power steam engines that drained mines and eventually drove machinery.



ADA LOVELACE

MECHANICAL CALCULATION

In the 1840s **Charles Babbage**, an English mathematician, worked on the development of an **Analytical Engine** which, had it been completed, would have been capable of performing calculations

similar to the early computers of the 1940s.

Ada Lovelace, a woman fascinated by mathematics, saw the possibilities of the machine. In 1843 she published the **first known computer program**.

Ingenious Inventions

The four essentials of technological advances are the right idea, the method to execute it, the perfect moment in time, and the availability of the right materials. The 19th century saw these elements merge in an explosion of technology, which introduced mass production and fed mass consumption.

The Industrial Revolution (see pp.292–95) had brought with it an increase in population and urbanization as well as an emerging “middle class,” which profited from industrial growth. Factories were filled with a labor force as agricultural manual work gave way to mechanization. Manufacturing produced cheaper goods for a mass market, encouraged by the rise in wages and standards of living. The traditional workplace, and the home, were being rapidly transformed.

Few men had a greater impact on life in the 19th century than Michael Faraday. Faraday discovered that an electric current was produced in a coil of wire when a magnet was moved through the coil.

His experiments with electricity in the 1830s led to dramatic developments in the sphere of communications. In 1837, British scientists William Cook and Charles Wheatstone patented the electric telegraph, a method of sending messages through wires to a remote receiver. Telegraphy expanded across the world. By 1876 the inventor Alexander Graham Bell, a Briton living

in the US who had spent much of his professional career working with the deaf, developed the telephone. This innovation went on to eclipse the telegraph.

Another electrical discovery in 1887 further revolutionized communications, when a German scientist named Heinrich Hertz proved the existence of radio waves. An Italian entrepreneur, Guglielmo Marconi, was convinced that Hertz’s work could lead to messages being sent over long distances without the use of cables and was soon transmitting “wireless” signals. In 1906 Marconi achieved his final goal by transmitting speech over the airwaves using radio transmissions.

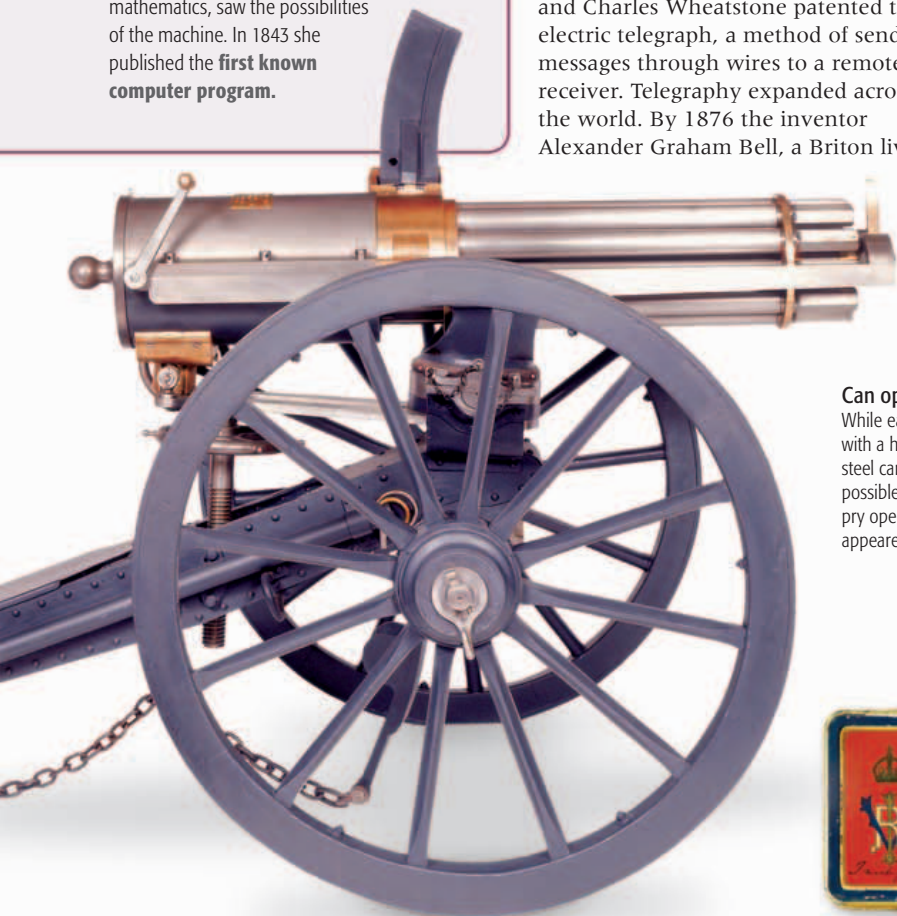
Other inventions were rapidly changing the world. By the turn of the 20th century wars were being

fought with more advanced weapons, food was preserved in new tin cans (see pp.290–91), and domestic refrigeration was evolving. Cars emerged on the roads, shipbuilding was prolific, railroads stretched across continents, and the first plane was about to take off. In the US, Wilbur and Orville Wright progressed by

way of gliders to a fabric-covered wooden biplane.

On December 17, 1903 they made a flight in the world’s first powered plane; by the early 1930s the modern airplane was beginning to take shape (see pp.458–61).

Domestic life was being slowly transformed by sewing machines, washing machines, electric stoves, electric heaters, and vacuum cleaners. Families had clearly defined “leisure time.” They listened to gramophones, which played flat discs, and, in March 1895, people thrilled to the sight of flickering pictures when a film of workers leaving the Lumière factory during their lunch hour was shown. Cinema was born.



Gatling gun

Richard Gatling’s mechanical gun of the 1860s consisted of six barrels mounted in a revolving frame. It was easy to use, reliable, and effective.

Can opener

While early tin cans were opened with a hammer and chisel, lighter steel cans made the can opener possible. “Bull’s head” pierce-and-pry openers, like this one, appeared in the 1860s.



Tin can

A discovery in 1810 that food could be preserved by heating it and sealing it in jars led two British men, Bryan Donkin and John Hall, to develop tin cans.



Berliner gramophone

Emile Berliner created the forerunner of modern records and record players. This 1890s model plays grooved flat discs, which were cheaply produced.



Bell “Box” telephone

Alexander Graham Bell had transmitted speech along a wire by 1876. This early Bell telephone has a combined trumpetlike mouthpiece and earpiece.